

2023 AMC 10A Problem 23 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 23

Problem:

If the positive integer c has positive integer divisors a and b with $c = ab$, then a and b are said to be complementary divisors of c . Suppose that N is a positive integer that has one complementary pair of divisors that differ by 20 and another pair of complementary divisors that differ by 23. What is the sum of the digits of N ?

Answer Choices:

- A. 9
- B. 13
- C. 15
- D. 17
- E. 19

Solution:

Write the factorizations as $a(a + 20) = b(b + 23) = N$. Thus $0 < b < a$. Multiply through by 4 and complete the square to see that

$$(2a + 20)^2 = (2b + 23)^2 - 129.$$

Thus

$$(2b + 23)^2 - (2a + 20)^2 = (2b + 2a + 43)(2b - 2a + 3) = 129 = 3 \cdot 43.$$

The positive divisors of 129 are 1, 3, 43, and 129. The number $2b + 2a + 43$ is a divisor of 129 that is greater than 43, so

$$2b + 2a + 43 = 129$$

and

$$2b - 2a + 3 = 1$$

This gives $a = 22$ and $b = 21$, so $a + 20 = 42$, $b + 23 = 44$, and hence $N = 22 \cdot 42 = 2^2 \cdot 3 \cdot 7 \cdot 11 = 21 \cdot 44 = 924$. The sum of its digits is $9 + 2 + 4 = 15$. (C) 15.

OR

As in the previous solution, if the factorizations are written as $a(a + 20) = b(b + 23) = N$, then $b < a$ and $a + 20 < b + 23$, from which $b < a < b + 3$.

Because a and b must be integers, a must equal $b + 1$ or $b + 2$. In the case when $a = b + 1$, $a(a + 20)$ becomes $(b + 1)(b + 21) = b^2 + 22b + 21$. Setting this equal to $b(b + 23) = b^2 + 23b$ yields the solution $b = 21$, which corresponds to $N = 21 \cdot 44 = 22 \cdot 42 = 924$, and the sum of the digits of 924 is 15. (C) 15.

In the case when $a = b + 2$, $a(a + 20)$ becomes $(b + 2)(b + 22) = b^2 + 24b + 44$. Setting this equal to $b(b + 23) = b^2 + 23b$ yields the only solution, $b = -44$. This does not meet the requirement that the complementary divisors be positive, although here as well $(-44) \cdot (-21) = (-42) \cdot (-21) = 924$.

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